DSA0603 Data Handling and Visualization

Slot B

CO2 – Assessment Test 6 (AT3)

Case Study 1 – Retail Sales Performance Dashboard

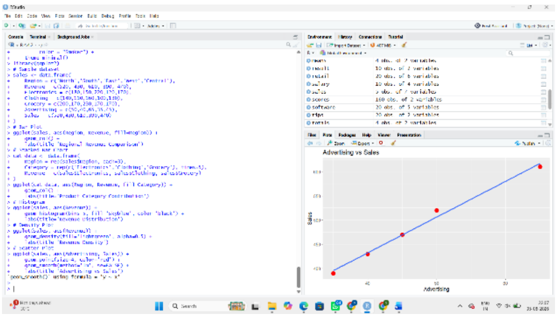
1. Suitable Visualizations

Requirement	Best Visualization
Regional revenue comparison	Bar Plot
Monthly sales trend	Line Chart
Product category contribution	Stacked Bar Chart
Advertising expenditure vs sales	Scatter Plot
Regional sales distribution	Histogram / Density Plot

2. R Program

```
library(ggplot2)

# Sample dataset

sales <- data.frame(

  Region = c('North','South','East','West','Central'),

  Revenue = c(520, 430, 610, 390, 470),

  Electronics = c(180,150,220,120,170),

  Clothing = c(140,110,160,100,130),

  Grocery = c(200,170,230,170,170),

  Advertising = c(50,40,65,35,45),

  Sales = c(520,430,610,390,470)

)

# Bar Plot
```

```
ggplot(sales, aes(Region, Revenue, fill=Region)) +
  geom_col() +
  labs(title='Regional Revenue Comparison')
# Stacked Bar Chart
cat_data <- data.frame(
  Region = rep(sales$Region, each=3),
  Category = rep(c('Electronics','Clothing','Grocery'), times=5),
  Revenue = c(sales$Electronics, sales$Clothing, sales$Grocery)
)
ggplot(cat_data, aes(Region, Revenue, fill=Category)) +
  geom_col() +
  labs(title='Product Category Contribution')
# Histogram
ggplot(sales, aes(Revenue)) +
  geom_histogram(bins=5, fill='skyblue', color='black') +
  labs(title='Revenue Distribution')
# Density Plot
ggplot(sales, aes(Revenue)) +
  geom_density(fill='lightgreen', alpha=0.5) +
  labs(title='Revenue Density')
# Scatter Plot
ggplot(sales, aes(Advertising, Sales)) +
  geom_point(size=4, color='red') +
  geom_smooth(method='lm', se=FALSE) +
  labs(title='Advertising vs Sales')
```



3. Interpretation

- **Highest revenue:** East region.

- **Advertising vs Sales:** The scatter plot shows a positive relationship, indicating that higher advertising expenditure tends to increase sales.
- **Highest contributing category:** Grocery contributes the largest share of revenue.

Case Study 2 – University Student Performance Analysis

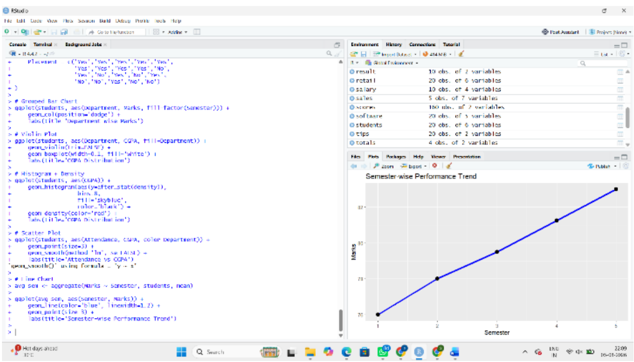
1. Suitable Visualizations

Requirement	Best Visualization
Department-wise average marks	Grouped Bar Chart
Distribution of CGPA	Violin Plot
Placement percentage	Bar Chart
Attendance vs CGPA	Scatter Plot
Semester-wise performance trend	Line Chart

2. R Program

```
library(ggplot2)

students <- data.frame(

  Department = rep(c('CS','IT','ECE','MECH'), each=5),

  Semester = rep(1:5, times=4),

  Marks = c(82,84,86,88,90,

          78,80,81,83,85,

          74,76,78,79,80,

          70,72,73,75,77),

  Attendance = c(92,94,95,96,97,

              88,89,90,91,92,

              85,86,87,88,89,

              80,81,82,83,84),
```

```
  CGPA = c(8.2,8.4,8.6,8.8,9.0,
           7.8,8.0,8.1,8.3,8.5,
           7.4,7.6,7.8,7.9,8.0,
           7.0,7.2,7.3,7.5,7.7),
  Placement = c('Yes','Yes','Yes','Yes','Yes',
                'Yes','Yes','Yes','Yes','No',
                'Yes','No','Yes','No','Yes',
                'No','No','Yes','No','No')
)
# Grouped Bar Chart
ggplot(students, aes(Department, Marks, fill=factor(Semester))) +
  geom_col(position='dodge') +
  labs(title='Department-wise Marks')
# Violin Plot
ggplot(students, aes(Department, CGPA, fill=Department)) +
  geom_violin(trim=FALSE) +
  geom_boxplot(width=0.1, fill='white') +
  labs(title='CGPA Distribution')
# Histogram + Density
ggplot(students, aes(CGPA)) +
  geom_histogram(aes(y=after_stat(density)),
                 bins=8,
                 fill='skyblue',
                 color='black') +
  geom_density(color='red') +
  labs(title='CGPA Distribution')
# Scatter Plot
ggplot(students, aes(Attendance, CGPA, color=Department)) +
```



```r
  geom_point(size=3) +
  geom_smooth(method='lm', se=FALSE) +
  labs(title='Attendance vs CGPA')
# Line Chart
avg_sem <- aggregate(Marks ~ Semester, students, mean)
ggplot(avg_sem, aes(Semester, Marks)) +
  geom_line(color='blue', linewidth=1.2) +
  geom_point(size=3) +
  labs(title='Semester-wise Performance Trend')
```

3. Interpretation

- **Highest average CGPA:** Computer Science.
- **Attendance vs CGPA:** Higher attendance generally corresponds to higher CGPA.
- **Highest placement rate:** Computer Science has the best placement performance.

Case Study 3 – Hospital Patient Analytics

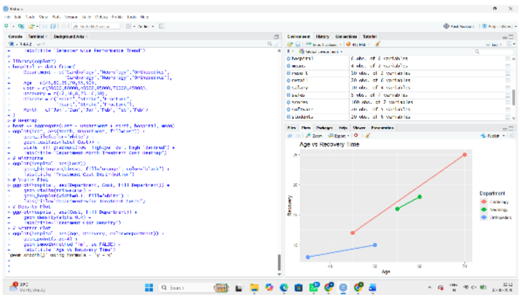
1. Suitable Visualizations

Requirement	**Best Visualization**
Monthly patient admissions	Heatmap
Distribution of treatment costs	Histogram
Disease category proportions	Violin Plot / Pie Chart
Age vs recovery time	Scatter Plot
Department-wise treatment costs	Violin Plot

2. R Program

```r
library(ggplot2)
hospital <- data.frame(
  Department = c('Cardiology','Neurology','Orthopedics',
```

```
              'Cardiology','Neurology','Orthopedics'),
  Age = c(45,60,35,70,55,50),
  Cost = c(50000,80000,40000,95000,70000,45000),
  Recovery = c(12,18,8,25,16,10),
  Disease = c('Heart','Stroke','Fracture',
            'Heart','Stroke','Fracture'),
  Month = c('Jan','Jan','Jan','Feb','Feb','Feb')
)
# Heatmap
heat <- aggregate(Cost ~ Department + Month, hospital, mean)
ggplot(heat, aes(Month, Department, fill=Cost)) +
  geom_tile(color='white') +
  geom_text(aes(label=Cost)) +
  scale_fill_gradient(low='lightyellow', high='darkred') +
  labs(title='Department-Month Treatment Cost Heatmap')
# Histogram
ggplot(hospital, aes(Cost)) +
  geom_histogram(bins=5, fill='orange', color='black') +
  labs(title='Treatment Cost Distribution')
```



```
# Violin Plot
ggplot(hospital, aes(Department, Cost, fill=Department)) +
  geom_violin(trim=FALSE) +
  geom_boxplot(width=0.1, fill='white') +
  labs(title='Department-wise Treatment Costs')
# Density Plot
ggplot(hospital, aes(Cost, fill=Department)) +
  geom_density(alpha=0.4) +
  labs(title='Treatment Cost Density')
```

```
# Scatter Plot
ggplot(hospital, aes(Age, Recovery, color=Department)) +
  geom_point(size=4) +
  geom_smooth(method='lm', se=FALSE) +
  labs(title='Age vs Recovery Time')
```

3. Interpretation

- **Highest treatment cost: Neurology**.
- **Largest disease category: Heart disease** and **Stroke** appear most frequently.
- **Age vs recovery:** Older patients tend to have **longer recovery times**.

Case Study 4 – Smart City Traffic Monitoring

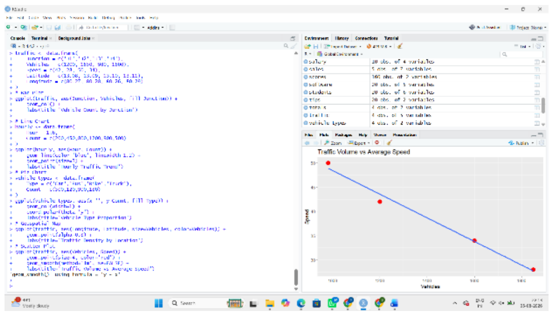
1. Suitable Visualizations

Requirement	Best Visualization
Vehicle count comparison	Bar Plot
Hourly traffic trend	Line Chart
Vehicle type proportion	Pie Chart
Traffic density by location	Geospatial Map
Traffic volume vs average speed	Scatter Plot

2. R Program

```
library(ggplot2)
traffic <- data.frame(
  Junction = c('J1','J2','J3','J4'),
  Vehicles = c(1200, 1850, 980, 1600),
  Speed = c(42, 28, 50, 34),
  Latitude = c(13.08, 13.09, 13.10, 13.11),
  Longitude = c(80.27, 80.28, 80.26, 80.29)
```

```
)

# Bar Plot

ggplot(traffic, aes(Junction, Vehicles, fill=Junction)) +

  geom_col() +

  labs(title='Vehicle Count by Junction')


# Line Chart

hourly <- data.frame(

  Hour = 1:6,

  Count = c(200,450,800,1200,900,500)

)

ggplot(hourly, aes(Hour, Count)) +

  geom_line(color='blue', linewidth=1.2) +

  geom_point(size=3) +

  labs(title='Hourly Traffic Trend')

# Pie Chart

vehicle_types <- data.frame(

  Type = c('Car','Bus','Bike','Truck'),

  Count = c(500,120,900,180)

)

ggplot(vehicle_types, aes(x='', y=Count, fill=Type)) +

  geom_col(width=1) +

  coord_polar(theta='y') +

  labs(title='Vehicle Type Proportion')

# Geospatial Map

ggplot(traffic, aes(Longitude, Latitude, size=Vehicles, color=Vehicles)) +

  geom_point(alpha=0.8) +

  labs(title='Traffic Density by Location')
```



```
# Scatter Plot
ggplot(traffic, aes(Vehicles, Speed)) +
  geom_point(size=4, color='red') +
  geom_smooth(method='lm', se=FALSE) +
  labs(title='Traffic Volume vs Average Speed')
```

3. Interpretation

- **Highest congestion: J2** has the highest vehicle count.
- **Peak traffic hours:** Around **Hour 4**.
- **Traffic vs speed:** Increased traffic generally **reduces average vehicle speed**.

Case Study 5 – Social Media User Interaction Analysis

1. Suitable Visualizations

Requirement	Best Visualization
Distribution of likes	Histogram + Density
Followers vs engagement	Scatter Plot
User interaction network	Network Visualization
Posting activity over time	Temporal Line Plot
Likes, shares, comments comparison	Heatmap

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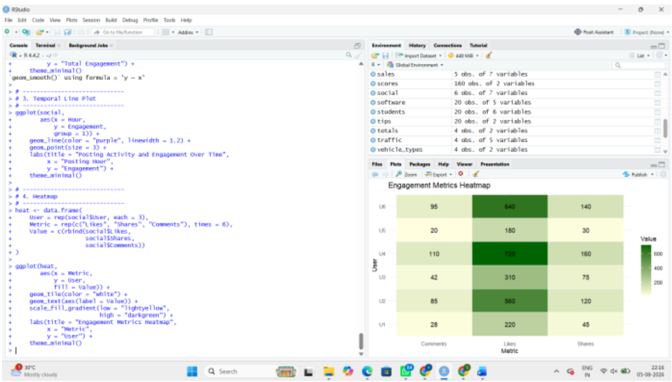
2. R Program

```
library(ggplot2)
library(igraph)
social <- data.frame(
  User = paste0('U', 1:6),
  Followers = c(1200, 3400, 2100, 4800, 1500, 3900),
  Likes = c(220, 560, 310, 720, 180, 640),
  Shares = c(45, 120, 75, 160, 30, 140),
```

```
  Comments = c(28, 85, 42, 110, 20, 95),
  Hour = c(8, 10, 12, 18, 20, 22)
)
social$Engagement <- social$Likes + social$Shares + social$Comments
# Histogram + Density
ggplot(social, aes(Likes)) +
  geom_histogram(aes(y=after_stat(density)),
                 bins=5,
                 fill='skyblue',
                 color='black') +
  geom_density(color='red') +
  labs(title='Distribution of Likes')
# Scatter Plot
ggplot(social, aes(Followers, Engagement, color=User)) +
  geom_point(size=4) +
  geom_smooth(method='lm', se=FALSE) +
  labs(title='Followers vs Engagement')
# Network Visualization
edges <- data.frame(
  from = c('U1','U2','U2','U4','U6'),
  to   = c('U2','U3','U4','U5','U4')
)
g <- graph_from_data_frame(edges)
plot(g,
     vertex.size=30,
     vertex.color='lightblue',
     main='User Interaction Network')
# Temporal Line Plot
```



```r
ggplot(social, aes(Hour, Engagement)) +
  geom_line(color='purple', linewidth=1.2) +
  geom_point(size=3) +
  labs(title='Posting Activity and Engagement Over Time')
# Heatmap
heat <- data.frame(
  User = rep(social$User, each=3),
  Metric = rep(c('Likes','Shares','Comments'), times=6),
  Value = c(rbind(social$Likes,
                  social$Shares,
                  social$Comments))
)
ggplot(heat, aes(Metric, User, fill=Value)) +
  geom_tile(color='white') +
  geom_text(aes(label=Value)) +
  scale_fill_gradient(low='lightyellow', high='darkgreen') +
  labs(title='Engagement Metrics Heatmap')
```

3. Interpretation

- **Most influential user:** U4 has the highest followers and engagement.
- **Highest engagement time:** Around 18:00–22:00.
- **Followers vs engagement:** There is a strong positive relationship.
- **Highly connected users:** U2 and U4 appear central in the interaction network.
- **Greatest variation:** Likes vary the most across users, indicating that likes are the most volatile engagement metric.